

Contents

1	Writing Conventions	2
1.1	Logical rules	2
1.2	Structure	2
1.3	Mathematical Conventions	4
1.4	Notation	4
2	To Do Liste	5
3	Differentiable structures on topological manifolds	6
4	Morse Theory	12
5	What are Spheres	22
6	Morse-Theoretic Atiyah-Hirzebruch Spectralsequence	26
6.1	Conley index	26
6.2	The Spectral sequence	29

1 Writing Conventions

In general I use British English.

1.1 Logical rules

Each item in a list will end with a point and start capitalized. If I list something where the order matters like:

1. Step one.
2. Step two.
3. Step three.

and for equivalent properties I use letters: For example, a Morse-Smale function f on a Riemannian manifold (M, g) satisfies:

- (a) f is smooth.
- (b) The critical points of f are non degenerate.
- (c) The (un-) stable manifolds intersect transversal.

1.2 Structure

In this thesis i have the following text-types

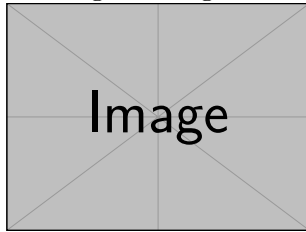
- (a) Free text that is not in any container.
- (b) The theorem container.
- (c) The lemma container.
- (d) The corollary container.
- (e) The proposition container.
- (f) The prove container.

Free Text is the main narrative of the thesis. It includes explanations, motivations, intuitions, examples, informal remarks, and transitions. Free text connects the formal content and helps guide the reader through the logical structure of the work.

Theorem 1.2.1 (An Example). *Used for major, central mathematical statements that are either original contributions or critical results from the literature. Theorems typically capture the most important claims and are often proved in detail.*

Proof Strategy:

Here I give the geometric idea of the proof and explain the needed steps.

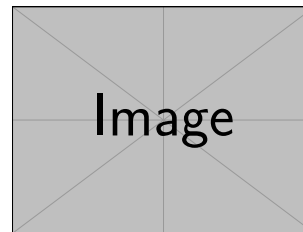


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Lemma 1.2.2 (An Example). *Used for technical results that support the proof of*

theorems or other key statements. Lemmas are usually not of independent interest but are essential building blocks within a logical argument.

Corollary 1.2.3 (An Example). Used for statements that follow immediately and straightforwardly from a previously proven theorem or lemma. Corollaries are usually included to highlight direct consequences of major results.

Proposition 1.2.4 (An Example). *Used for results that are mathematically relevant but not as central as theorems. Propositions often encapsulate useful facts, structural properties, or classifications that are too substantial for a lemma but not significant enough for a theorem.*

Proof. This container is pretty obvious □

1.3 Mathematical Conventions

Elements in a vector bundle will have greek letters. For example $\xi \in TM$. For restrictions I will use the notation $f|_A$ that is written as :

`\restr{f}{A}`.

In the realm of differential geometrie i will **always** use lower indices to denote components. I will **not** use the einstein summation, since it really appears and I never reallyly deal with cumbersome computations (in this context).

1.4 Notation

Symbol	Description	Context
A^t	The transpose of the Matrix A	Linear algebra
M	Smooth manifold	Differential geometry
$\dim M$	Dimension of manifold M	Differential geometry
TM	Tangent bundle of M	Differential geometry
$T_p M$	Tangent space at point p	Differential geometry
g	Riemannian metric	Riemannian geometry
$\text{Hess}_p(f)$	Hessian of f in p	Morse theory
$M_p(f)$	The matrix correspoding to the Hessian. The used chart is suppressed	Morse theory
$\text{Crit}_k(f)$	Set of critical points of index k	Morse theory
λ_p or λ_p^f	Index of the critical point p	Morse theory

Classical and often used notation for maps:

Symbol	Description
$i : A \rightarrow B$	If $A \subseteq B$ this will always denote the inclusion
$B_{\mathfrak{b}} : \mathbb{R}^n \rightarrow V$	Isomorphism for the Basis $\mathfrak{b} = (v_i)_i$ of V : $(a_1, \dots, a_n) \mapsto \sum a_i v_i$
$q_{\mathfrak{b}} : V \rightarrow \mathbb{R}^n$	Coordinate map: $v = \lambda_i v_i \mapsto (\lambda_i)_i$ and also $B_{\mathfrak{b}}^{-1}$

2 To Do Liste

- How exactly is the morse boundary operator gievn? Make use of the map E_q^u and J .
- tripele boundary operator.
- wedge produkt and smash produkt commute...

3 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of interest for mathematicians and physicists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 3.0.1 (Topological Manifold). A second countable Hausdorff space M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p , an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : U \rightarrow V$ that is a homeomorphism. We call the map $\varphi : U \rightarrow V$ a **chart around p on M** and φ^{-1} a **local coordinate system around p on M** .

Definition 3.0.2 (Differentiable Manifold). Let M be a topological manifold of dimension m .

1. A **differentiable atlas of class $r \in \mathbb{N} \cup \{\infty\}$** is a family of charts $\mathfrak{A} = (\varphi_i : U_i \rightarrow V_i)_{i \in I}$ such that
 - a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M .
 - b) For every pair $(i, j) \in I^2$ the **transition function**:

$$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\rightarrow \varphi_i(U_i \cap U_j) \\ x &\mapsto (\varphi_i \circ \varphi_j^{-1})(x) \end{aligned}$$

is differentiable of class r .

We call such an atlas a C^r -atlas.

2. Two C^r -atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r -atlas.

A **differentiable structure of class r on M** is an equivalence class c of C^r -atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 3.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$.

Definition 3.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \rightarrow \mathbb{R}$$

differentiable of class r , if for any one (and hence for all) $(\varphi_i)_{i \in I} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r . For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in C(U, \mathbb{R}) \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 3.0.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$ -algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$ -algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \text{res}_V^U : \mathcal{E}(U) &\rightarrow \mathcal{E}(V) \\ f &\mapsto f|_V . \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. $\square$

Definition 3.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p , if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p . We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{ open}, p \in U} \mathcal{E}(U) \right) / \sim_p .$$

For the set of germs and call it the **stalk in** p . Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 3.0.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$ -algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \text{eval}_p : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\text{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\text{eval}_p)$. $\square$

Definition 3.0.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$ -linear map $\delta : \mathcal{E}_p(M) \rightarrow \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f, g \in \mathcal{E}_p(M) .$$

We call $\text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R}) .$$

Corollary 3.0.9. Let (M, c) be a smooth manifold and $\varphi : U \rightarrow V$ be a chart around p with $x_0 = \varphi(p)$ ($\varphi \in \mathfrak{A} \in c$). Then

$$\xi = \frac{\partial}{\partial x_j} \Big|_p : \mathcal{E}_p(M) \rightarrow \mathbb{R}$$

$$f_p \mapsto \xi(f_p) = \frac{\partial}{\partial x_j} \Big|_{x_0} (f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p \right)$$

defines a basis of TM_p . Hence, the dimension of TM_p is m .

Definition 3.0.10. For a smooth manifold (M, c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \rightarrow M$$

$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \rightarrow V$

$$\frac{\partial}{\partial x_j} : U \rightarrow \pi^{-1}(U)$$

$$p \mapsto \frac{\partial}{\partial x_j} \Big|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m.$$

We can induce a topology on TM such that all those trivializations are continuous. This then gives an atlas for TM such that we have a $2m$ -dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \rightarrow \mathbb{R}^m \times \mathbb{R}^m; x \mapsto (\pi(x), q_{\varphi \circ \pi}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p)$ that depends on the chart φ_i . In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m . We call TM the **tangent bundle**.

Definition 3.0.11 (The Derivative). Let (M, c) and (M', c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \rightarrow M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \rightarrow V'$$

are smooth. A given smooth function induces a smooth function between the Tangent bundles as follows:

$$Df : TM \rightarrow TM', Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_p M$, $g_p \in E_p(M')$.

Corollary 3.0.12. Let $f : M \rightarrow \mathbb{R}$ be smooth and $\varphi : U \rightarrow V$ be a chart. Then we can interpret df as a one-form. To be precise assume q to be the coordinate function $T\mathbb{R} \rightarrow \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ df(v) = v \cdot f$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notation so vector fields can take functions as an input.

Proof. We prove this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M, \mathbb{R})$ vectorspace. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$df\left(\frac{\partial}{\partial x_i}\Big|_p\right)(g_p) = \frac{\partial}{\partial x_i}\Big|_p(g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0}(g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p(g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p(f_p)$$

Hence, $df(v) = v(f) \frac{\partial}{\partial x}$ letting us conclude the statement. $\square$

Definition 3.0.13 (A Metric). Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual Tangent bundle with itself is a **riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_p M$ $g_p(v_p, w_p) = 0 \ \forall w_p \in T_p M$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p) \ \forall v_p, w_p \in T_p M$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0 \ \forall v_p$ and vanishes only for $v_p = 0$.

We call a tuple (M, g) a **riemannian manifold**.

Definition 3.0.14. Let M, N, L be smooth manifolds and $f : M \times N \rightarrow M' \times N'$. Let $X \in \Gamma(TM)$ and $Y \in \Gamma(TN)$. We define $\tilde{X} \in \Gamma(T(M \times N))$ as follows. Let $f_p \in \mathcal{E}_p(M, N)$

$$\tilde{X}(f_p) = X((f \circ i_M)_{p_M}) \quad \text{where } p_M = \pi_1(p).$$

Definition 3.0.15 (The Gradient). Assume that (M, g) is a smooth manifold together with a riemannian metric. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\text{grad}(f) \in \Gamma(TM)$ such that for any vector field V :

$$q \circ df(V) = g(\text{grad}(f), V).$$

Here q denotes the canonical coordinate function $T\mathbb{R} \rightarrow \mathbb{R}$.

Definition 3.0.16. Let (M, g) be a riemannian manifold and $\varphi : U \rightarrow V$ be a chart. We define the smooth functions $g_{ij} : U \rightarrow \mathbb{R}$ to be :

$$g_{ij} = g\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right).$$

Then if two vectorfields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v, w) = \sum_{ij} g_{ij} v^i w^j : U \rightarrow \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via crammers rule) to get smooth functinos

$$g^{ij} : U \rightarrow \mathbb{R}$$

$$p \mapsto g^{ij}(p) = ((g_{kl}(p))_{kl}^{-1})_{ij}$$

Lemma 3.0.17. *Let $\varphi : U \rightarrow V$ be a chart. Then the gradient has the local form:*

$$\text{grad}(f) = \sum_{i,j} g^{ij} \left(\frac{\partial}{\partial x_i}(f) \right) \frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij} \frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j}.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum w^i \frac{\partial}{\partial x_i}$. Then $g(v, w) = \sum_{i,j} g_{ij} v^i w^j$. Suppose that $\text{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \rightarrow \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ df \left(\frac{\partial}{\partial x_j} \right) = q \circ \frac{\partial f}{\partial x_j} = g(\text{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$(G^1, \dots, G^m)(g_{ij}) = \left(\frac{\partial}{\partial x_1}(f), \dots, \frac{\partial}{\partial x_m}(f) \right)$$

$$\Leftrightarrow (G^1, \dots, G^m) = \left(\frac{\partial}{\partial x_1}(f), \dots, \frac{\partial}{\partial x_m}(f) \right) (g^{ij}).$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$. □

Definition 3.0.18 (A Dynamical System). Let M be a smooth manifold. A **dynamical system on M** is given by a smooth map $\varphi : \Omega \rightarrow M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t, p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t, p))) = \varphi(s + t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi : t(p) := \varphi(t, p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 3.0.19. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \rightarrow \text{Diff}, \quad t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t .

Definition 3.0.20 (Vector Field of a Dynamical System). Let φ be a dynamical system on a smooth manifold M . Then we call the mapping

$$p \mapsto X(p) := \left. \frac{d}{dt} \right|_{t=0} \varphi_t(p) \in T_p M$$

the **associated vector field**. This is in fact a smooth vector field.

Whilst differentiating leads to a smooth vector field, we can also go the other direction by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words

Theorem 3.0.21. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

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werden

4 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [banyaga2004lectures]. In Morse Theory, we will always deal with a smooth function $f : M \rightarrow \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 4.0.1 (Kritical Points). We call a point $p \in M$ **critical**, if $df|_p$ vanishes meaning that

$$df|_p : T_p M \rightarrow T_{f(p)} \mathbb{R}$$

is trivial. By $\text{Crit}(f)$ we denote the **set of critical points**.

Definition 4.0.2 (The Hessian). Let $f : M \rightarrow \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \text{Crit}(f)$ is a symmetric bilinear function

$$\text{Hess}_p(f) : T_p M \times T_p M \rightarrow \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_p M$ be two tangend vectors. Now choos two vectorfields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\text{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))(p)$$

Compare 3.0.12 for the notation.

Definition 4.0.3. A smooth function $f : M \rightarrow \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a Symmetric bilinear and non-degnereate Map. For such a map there exists the following theorem:

Theorem 4.0.4 (Sylvesters Law of Inertia). *Let $A \in \text{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \text{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k', l = l'$ and $\text{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \text{rank}(T^t \circ A \circ T) = \text{rank}(A) = \text{rank}(T'^t \circ A \circ T') = k' + l'. \quad (1)$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max \{ \dim(U) \mid U \subseteq \mathbb{R}^n \text{ subspace such that } x^t A x > 0 \ \forall x \in U \setminus \{0\} \}$$

So we start by showing „ $\leq$ “: Denote the first k colomus of T with $x_1, \dots, x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda_i x_i$ we have that by bilinearity

$$x^t A x = \sum_{i=1}^k \lambda_i x_i^t A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^t A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0. \quad (2)$$

This concludes the first inequality.

Now let U be any k -dimensional subspace such that for all non zero $x \in U$ $x^t A x > 0$. By a calculation analog to the one above we have that for any $x \in W := \text{span}(x_{k+1}, \dots, x_n)$ the number $x^t A x$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k. \quad (3)$$

This is the last inequality proving the statement. $\square$

Corollary 4.0.5. Let $\varphi : U \rightarrow V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

Proof. This is just an easy calculation, since $\text{Hess}_p(f) \left(\frac{\partial}{\partial x_i} \Big|_p, \frac{\partial}{\partial x_j} \Big|_p \right)$ is given by

$$\left(\frac{\partial}{\partial x_i} \Big|_p \left(\frac{\partial}{\partial x_j} (f) \right) \Big|_p \right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)$$

$\square$

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \rightarrow V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function. If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transforms from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = d\varphi_{12}^t \cdot M_p^2(f) \cdot d\varphi_{12}$$

Hence we can define the following:

Definition 4.0.6. By Sylvesters Law of inertia, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p , denoted with λ_p . If we cannot suppress the function without loss of clarity we will write λ_p^f . We denote the set of all critical points of index k by $\text{Crit}_k(f)$.

Theorem 4.0.7 (Morse Lemma). *Let p be a critical point of index k of the Morse funktion $f : M \rightarrow \mathbb{R}$. There exists a centered chart $\varphi : U \rightarrow \mathbb{R}^m$ around p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^k -x_i^2 + \sum_{i=k+1}^m x_i^2$$

Proof. Compar for example Lemma3.11 in [banyaga2004lectures]. $\square$

hier noch
mehr
Quellen!

Definition 4.0.8. Let $\varphi_t : M \rightarrow M$ be the local one-parameter group of diffeomorphisms generated by $-\text{grad}(f)$, i.e. if $e : \mathbb{R} \rightarrow T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{d}{dt} \varphi_t(x) \circ e = -\text{grad}(f)(\varphi_t(x)), \quad \text{and} \\ \varphi_0(x) = x.$$

We call that group the **gradient vector field** of f with respect to the gradient g . The integral curve $\gamma_x : (a, b) \rightarrow M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a **gradient flow line**.

Next we will deepen our **local** understanding of said vector fiel. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can finde one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 4.0.9. *Assume that $\psi : U \rightarrow V$ is a chart and $L : \mathbb{R}^m \rightarrow \mathbb{R}^m$ a linear function. Denote the local coordinate maps of the Tangend bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ . Then the diagramm*

$$\begin{array}{ccc} T_p M & \xrightarrow{q_\psi|_p} & \mathbb{R}^m \\ & \searrow q_{L \circ \psi}|_p & \downarrow L \\ & & \mathbb{R}^m \end{array}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi \circ L}|_p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x}_0$, $(v_1, \dots, v_m) = v \in \mathbb{R}^m$ and denote the basis of $T_p M$ coming from

the chart ψ by $\left(\frac{\partial}{\partial x_i}\bigg|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\bigg|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
 \left[(q_{\psi \circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\bigg|_p(f_p) \\
 &= \sum_i v_i \frac{\partial}{\partial x_i}\bigg|_{\tilde{x}_0}(f \circ \psi^{-1} \circ L^{-1}) \\
 &= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\bigg|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\bigg|_{\tilde{x}_0} \\
 &= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\bigg|_p(f_p) \cdot \lambda_{ji} \\
 &= \sum_{i,j} v_i q_{\psi}\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
 &= \left[q_{\psi}\big|_p^{-1}\left(\sum_{i,j} v_i \lambda_{ji} e^j\right)\right](f_p) \\
 &= \left[q_{\psi}\big|_p^{-1}(L^{-1}(v))\right](f_p)
 \end{aligned}$$

This concludes the proof. $\square$

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 4.0.10 (Simultaneous Diagonalisation of Quadratic Forms). *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on $T_p M$. Then there exists a Morse chart ψ around p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a diagonal matrix $\text{diag}(\mu_1, \dots, \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precise this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_{\psi}\big|_p \oplus q_{\psi}\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda Gv$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with

respect to the bilinear form defined by G , such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (H v_j)^T v_i = \lambda_j (G v_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i).$$

Hence we have the equality

$$\lambda_j (G v_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1 | \dots | v_m]$ be the matrix whose columns are these G -orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z . In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H(Lz) = z^T L^T H L z \end{aligned}$$

Since $H v_i = \lambda_i G v_i$, we have $L^T H L = \text{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive). This is true to Sylvester's law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\text{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \text{diag}(\mu_1, \dots, \mu_m)$ with $\mu_j > 0 \ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y)) = \sum_{i=1}^l -y_i^2 + \sum_{i=l+1}^m y_i^2$$

Furthermore, we want to inspect the metric $g|_p : T_p M^2 \rightarrow \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$ -induced basis $\left(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p \right)$. By the lemma 4.0.9

$$g(q_{L^{-1} \circ \psi}^{-1} \Big|_p, q_{L^{-1} \circ \psi}^{-1} \Big|_p)(v, w) = g(q_\psi^{-1} \Big|_p, q_\psi^{-1} \Big|_p)(Lv, Lw) = (Lv)^T G L w = v^T L^T G L w$$

□

Definition 4.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulas around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 4.0.12 (Jacobian of the Gradient). The gradient is a section into the tangent bundle, $\text{grad}(f) : M \rightarrow TM$. Let $\psi : M \supseteq U \rightarrow V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_p M$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\text{grad}(f))_\psi(x) := J(q_\psi \circ \text{grad}(f) \circ \psi^{-1})(x) = \left(\frac{\partial(q_\psi \circ \text{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x) \right)_{ij}$$

Lemma 4.0.13. Let ψ be a coordinate system around a critical point p . I.e. $\psi : p \in U \rightarrow V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of $T_x M$. Then the Jacobian of the gradient reads

$$J(\text{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right)_{ij}. \quad (4)$$

Proof. Let q_ψ denote the coordinate funktion $TU \rightarrow \mathbb{R}^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \text{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k} \right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} q_\psi \circ \text{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k} \right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}. \end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f) \circ \psi^{-1}\big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}.$$

□

Now that we now how the gradient locally looks like it is reasonable to assume, that the flow locally looks like its exponential:

Lemma 4.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \rightarrow M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $d\varphi_t|_p : T_p M \rightarrow T_{\varphi_t(p)} M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp \left(- \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right)_{ij} \cdot t \right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \rightarrow U$. For fixed x we can calculate

$$q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) = -q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x} \frac{\partial}{\partial t} \psi \circ \varphi_t(\psi^{-1}(x)) \\ &= \frac{\partial}{\partial x} q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x))) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f)(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))) \\ &= -\frac{\partial}{\partial x} (q_\psi \circ \text{grad}(f) \circ \psi^{-1}) \frac{\partial}{\partial x} (\psi \circ \varphi_t(\psi^{-1}(x))) \end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x} \psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp \left(-\frac{\partial}{\partial x} (q_\psi \circ \text{grad}(f) \circ \psi^{-1}) \cdot t \right).$$

The preceding lemma now finishes the proof. □

Hence in a refined Morse chart we have:

Remark 4.0.15. Let ψ be a refined Morse chart, and hence:

- (a) $g^{ik}(0) = \delta_{ik} \mu_k$ with $\mu_k > 0$ for any k .
- (b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned} \frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\text{grad}(f))(p) \cdot t) \\ &= \exp \left(- \sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k} (0) \underbrace{\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}} \cdot t \right) \\ &= \text{diag} \left(e^{2\mu_1 t}, \dots, e^{2\mu_{\lambda_p} t}, -e^{2\mu_{\lambda_p+1} t}, \dots, -e^{2\mu_m t} \right) \end{aligned}$$

we now skip a lot of theory that essentially uses the (un-)stable manifold theorem and the λ -lemma, to give the definition of the Morse chain complex for a Morse datum.

Definition 4.0.16 (Morse Homology). Let $f : M \rightarrow \mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M, g) . Furthermore let $\mathfrak{o}$ be an choice of orientations for all unstable Tangendspaces $(T_q^u M$ for all $q \in \text{Crit}(f)$). The tuple $\mathfrak{A} := (g, f, \mathfrak{o})$ is a **Morse datum**. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ) :

$$C_k(M, \mathfrak{A}, \Lambda) := \bigoplus_{\text{Crit}(f)} \Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q \rightarrow p)$. For each $\gamma \in \mathcal{M}(q \rightarrow p)$ we assign a sign $n_\gamma \in \{\pm 1\}$. Let $x \in \gamma$. Then the tangent space in x splits $T_x W(q \rightarrow) = N_x(\gamma) \oplus \langle -\text{grad}(f)(x) \rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to $T_p^u M$. Hence, the orientation in $T_q^u M$ can be compared to the one in $T_p^u M$ and the sign n_γ can be chosen accordingly. The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q) := \sum_{p \in \text{Crit}_k(f)} \sum_{\gamma \in \mathcal{M}(q \rightarrow p)} n_\gamma$$

This gives rise to a chain complex and hence gives rise to a homology theory, and a cohomology theory by applying the hom functor to the chain complex.

Lemma 4.0.17 (Orientation and Exact Sequences). *As stated before, an orientation of a Vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1, \dots, b_m)$ of V and hence the produkt $b_1 \wedge \dots \wedge b_m$ is an element of the maximal exterior product $\det(V) = \Lambda^m V$. Two ordered bases define different elements in $\det(V)$ where there difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence an orientation is a choice of one connected component of $\det(V) \setminus \{0\}$. Now a short exact sequence of vector spaces*

induces a canonical isomorphism

$$\det(A) \otimes \det(C) \rightarrow \det(B)$$

$$(a_1 \wedge \cdots \wedge a_l) \otimes (c_1 \wedge \cdots \wedge c_k) \mapsto (f(a_1) \wedge \cdots \wedge f(a_l) \wedge \tilde{c}_1 \wedge \cdots \wedge \tilde{c}_k),$$

where $\tilde{c}_j \in g^{-1}(c_j)$. The well definition of this map (invariance under choosen lift of c) can be calculated, since two different choices differ in an element in the kernal of g which equals the image of f and hence the different choices vanish in the wedge produkt. By this isomorphism we get a orientation induced in the middle from the outside.

Remark 4.0.18 (The Counting of Flow Lines). There are several ways to formulate the counting of flow lines that happens in the boundary operator. Let $\gamma_j \in \mathcal{M}(q \rightarrow p)$ and $x_j \in \gamma_j$. Instead of using the parallel transport, we can introduce two different ordered bases in $T_{x_j}W(q \rightarrow)$. We use the following notation: Let $\gamma_j \in \mathcal{M}(q \rightarrow p)$ and $x_j \in \gamma_j$. For this we define $N_{x_j}^{W(q \rightarrow)}\gamma_j$ be the normal bundle of γ_j as a submanifold of $W(q \rightarrow)$. Since γ_j is contractible we can trivialize said normal bundle to get an bundle isomorphism:

$$\nu_j : N_{x_j}^{W(q \rightarrow)}\gamma_j \rightarrow \gamma_j \times T_p^u M.$$

Let $\nu_j|_{x_j}$ denote the isomorphism $N_{x_j}^{W(q \rightarrow)}\gamma_j \rightarrow T_p^u M$. An orientation of $T_p^u M$ induces then one on $N_{x_j}^{W(q \rightarrow)}\gamma_j$ and we define the bases $(-\text{grad}_{x_j}(f))$ to be positive. Hence we get the exact sequence

$$0 \longrightarrow \langle -\text{grad}_{x_j}(f) \rangle \longrightarrow T_{x_j}W(q \rightarrow) \longrightarrow N_{x_j}^{W(q \rightarrow)}\gamma_j \longrightarrow 0$$

$$\uparrow (v_j|_{x_j})^{-1}$$

$$T_p^u M$$

That introduces an orientation in the middle. Another way to get an orientation there is to trivialize the bundle via the bundle isomorphism:

$$\mu : TW(q \rightarrow) \rightarrow W(q \rightarrow) \times T_p^u M.$$

This trivialization together with an orientation on $T_p^u M$ orients the middle in the above exact sequence. Now the deciding factor that gives γ_j its sign is whether they agree.

Here i write a little bit about the Morse cohomology in german

Theorem 4.0.19 (Morse Cohomology). *Um die Morse Kohomologie zu erhalten, dualisieren wir den Kettenkomplex für einen Ring G mit 1. Sei $\text{Crit}_k = p_1, \dots, p_l$*

eine Basis der k -ten Kettengruppe. Für

$$\eta_i : C_k \rightarrow G$$

$$p_j \mapsto \begin{cases} 0 & \text{if } j = i \\ 1 & \text{else} \end{cases}$$

bekommen wir eine Basis $(\eta_1, \dots, \eta_l)$ der Cocetten Gruppe. Die Randabbildung ist nun der spannende Teil. Seien $(q_1, \dots, q_r)$ kritische Punkte von Index $k+1$ und $(\xi_j)_{j=1, \dots, r}$ die resultierende Basis von C^{k+1} . Dann

$$d_k : C^k \rightarrow C^{k+1}$$

$$\eta_i \mapsto d\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_f(q_j, p_i) \xi_j.$$

Hier beschreibt $n_f(q, p)$ das zählen von Flusslinien von q nach p mittels f . Indem wir kritische Punkte und die Formen identifizieren erhalten wir das folgende:

$$C^k = \bigoplus_{q \in \text{Crit}_k} \mathbb{Z} = C_k.$$

$$d : C^k \rightarrow C^{k+1}$$

$$q \mapsto \sum_{p \in \text{Crit}_{k+1}} n_f(q, p) = \sum_{p \in \text{Crit}_{k+1}} n_{-f}(p, q) = \partial_{-f}.$$

Hier bezeichnet ∂_{-f} den Randoperator $C(M, -f, g)$.

Theorem 4.0.20 (Poincaré Dualität). Schnell sehen wir, dass

$$C^k(M, f) = C_{m-k}(M, -f)$$

damit erhalten wir:

$$H^k(M) \cong H^k(M, f) = \ker(\partial_{-f}) / \text{im} \partial_{-f} = H_{m-k}(M, -f) \cong H_k(M)$$

5 What are Spheres

In this section we want to inspect representations of homotopical Spheres. To give a little overview we will define the sphere to come with an orientation. To be more specific we have the definition:

Definition 5.0.1 (The k -Sphere). Let $\mathbb{R}^k$ be the k -dimensional vector space given as the span of $e_1, \dots, e_k$ together with the eucliden metric $\|\cdot\|_2$. Then the set

$$S^k := \{x \in \mathbb{R}^k \mid \|x\|_2 = 1\}$$

is called the k -sphere.

This sphere will be our save haven where we will always come back to. In this definition there is no room for "different orientations on a sphere". In our reality we will detect the two "different orientations" of a space A homotopic to a sphere that arise from two constructions as follows: assume that α and β are the maps $A \rightarrow S^k$. then there is an endomorphism $f : S^k \rightarrow S^k$ such that our diagramm commutes:

$$\begin{array}{ccc} & A & \\ \alpha \swarrow & & \searrow \beta \\ S^k & \xrightarrow{f} & S^k \end{array}$$

Now $f^* : K^n(S^k) \rightarrow K^n(S^k)$ is an isomorphism and hence either the identity or minus the identity.

Now we have a lot of spaces that are homotopic or even homeomorph to the sphere. Just to name a view constructions:

1. The disc in $\mathbb{R}^k$ modulo its boundary $D^k/\partial D^k$.
2. The cone over the boundary of a D^k disc.
3. The resduced and unreduced susopension of a $k-1$ sphere.
4. The one-point compactification of a vector space together with an oriented basis.

We already saw how we can map the one-point compactification of a vector space with an ordered basis to a sphere via the stereographic projections from the south pole. So lets inspect the other construction

Corollary 5.0.2. Given the disk $D^k \subseteq \mathbb{R}^k$ we have a homeomorphism

$$\overline{D^k}/\partial D^k \rightarrow S^k$$

Proof. We will construct a proper map from the open disk to R^k and by this we get a homeomorphism between their one-point-compaktifications.

$$\begin{aligned} s : D^k &\rightarrow \mathbb{R}^k \\ x &\mapsto x \cdot \frac{1}{1 - \|x\|_2} \end{aligned}$$

This is a homeomorphism and proper. Hence, we get a map between the onepoint compactifications

$$\overline{D^k}/\partial D^k \rightarrow \mathbb{R}^k \cup \{\infty\}$$

and the latter is homeomorphic to the Sphere. $\square$

Corollary 5.0.3. Let $\overline{D^k} \cup C\partial D^k$ be the cone over the boundary of the disk. This is homeomorphic to the k -sphere.

Proof. Again we will use the language of onepoint compactifications. First we give a homeomorphism from the "open cone" over the boundary to the open disc. To do this we include our space into R^k as follows:

$$f : \overline{D^k} \cup \partial D^k \times [0, 1) \rightarrow R^k$$

$$x \mapsto \begin{cases} x & \text{if } x \in \overline{D^k}, \\ (1+t)x & \text{if } (x, t) \in \partial D^k \times [0, 1). \end{cases}$$

This is a homeomorphism (if we restrict the image to $\{x \in R^k \mid \|x\|_2 < 2\}$). This can be seen as follows: Obviously it is bijective and the continuity can be derived from the gluing lemma. To see the continuity of the inverse we can explicitly describe it:

$$f^{-1} : \{x \in R^k \mid \|x\|_2 < 2\} \rightarrow \overline{D^k} \cup \partial D^k \times [0, 1)$$

$$x \mapsto \begin{cases} x & \text{if } x \in \overline{D^k}, \\ (\frac{x}{\|x\|_2}, (1 - \|x\|_2)) & \text{if } \|x\|_2 \geq 1, \end{cases}$$

Again, gluing lemma gives continuity, and a calculating gives us that they are inverse to each other. To see that the map is proper, we check if preimages of compact sets are compact. So let $K \subset \{x \in R^k \mid \|x\|_2 < 2\}$ be compact. Then $K = A \cup B$ where $A = K \cap \overline{D^k}$ and $B = K \cap \{x \in R^k \mid 1 \leq \|x\|_2 < 2\}$. Now A and B are again compact. (A is the intersection of two compact Hausdorff spaces and B can also be constructed as such a intersection) Since $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ we have to check if f^{-1} is a closed map restricted to $\overline{D^k}$ and $\{x \in R^k \mid 1 \leq \|x\|_2 < 2\}$. But those restrictions give rise to homeomorphisms and hence they are closed. Now we have a proper homeomorphism which lets us conclude, that the map induced on their onepoint compactifications are also homeomorphisms. After the rescaling:

$$r : \{x \in R^k \mid \|x\|_2 < 2\} \rightarrow D^k$$

$$x \mapsto \frac{x}{2}$$

we have a proper homeomorphism $\overline{D^k} \cup \partial D^k \times [0, 1) \rightarrow D^k$. This lets us deduce the statment, as the onepoint compactification of the disk can be mapped to the sphere as seen in the corollary above. $\square$

We will later use similar maps in our main prove, but now we want to talk about spheres in K -theorie.

Definition 5.0.4. Let $f : S^k \rightarrow S^k$ be continouos. We know that $f^* : K^k[S^k] \rightarrow K^k[S^k]$ is a group homomorphism and via the map $\mathbb{Z} \rightarrow K^k[S^k]$ induced from mapping $1 \mapsto \beta$ we get a induced map $\tilde{f}$ such that the diagramm koomutes:

$$\begin{array}{ccc} K^k[S^k] & \xrightarrow{f^*} & K^k[S^k] \\ \uparrow & & \uparrow \\ \mathbb{Z} & \xrightarrow{\tilde{f}} & \mathbb{Z} \end{array}$$

Hence $\tilde{f}$ is the multiplication with a number $d \in \mathbb{Z}$. This number is obviously homotopie invariant and hence we can define the degree map:

$$\begin{aligned} \deg : \pi_k(S^k) &\rightarrow \mathbb{Z} \\ [f] &\mapsto d \end{aligned}$$

Corollary 5.0.5. The map $\deg$ is an group isomorphism.

Proof. We start by proving that it is a group homomorphism. Let f, g be representatives of two elements in $\pi_k(S^k)$. Let $\deg(f) = m$, and $\deg(g) = n$. The map $f + g$ is defined as the composition:

$$S^k \xrightarrow{c} S^k \wedge S^k \xrightarrow{f \wedge g} S^k$$

Here c denotes the collaps of the equator. □

Corollary 5.0.6. We want to understand homöomorphisms between boukettes of spheres. Or to beginn with let:

$$f : S^k \rightarrow \bigvee S^k$$

Now how does f^* look like? for this we

Proof. L □

Corollary 5.0.7. If we collaps a cone $C(S^k)$ over a shpere at its base, we obtain a sphere.

Proof. We define the homeomorphism as follows:

$$\begin{aligned} \Omega : C(S^k)/\{0\} \times S^k &\rightarrow S^{k+1} \subset \mathbb{R}^{k+2} \\ \overline{(t, x)} &\mapsto (-\cos(\pi t), \sin(\pi t)x). \end{aligned}$$

This homeomorphism has the following properties: If $t > \frac{1}{2}$ we end up in the upper hemisphere. Furthermore, we can view the domain and its image as topological manifolds and if we cut out the collapsed points, we even have a smooth manifold. Here the map is differentiable. □

We can use this map to construt a new map form $\overline{D^k}/\partial D^k \rightarrow S^k$ as follows:

Definition 5.0.8. We can identify $\overline{D^k}/\partial D^k$ with $C(S^{k-1})/\{0\} \times S^{k-1}$ as follows:

$$s : \overline{D^k}/\partial D^k \rightarrow C(S^{k-1})/\{0\} \times S^{k-1}$$

$$\bar{x} \mapsto \left(\|x\|, \frac{x}{\|x\|} \right).$$

The composition $\Gamma := \Omega \circ s$ now gives a map

$$\overline{D^k}/\partial D^k \rightarrow S^k$$

6 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

6.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 6.1.1 (Compact invariant isolated subset). For a flow $\varphi_t : M \rightarrow M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 6.1.2 (Index pairs). Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of S**, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (N \setminus L)$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in N**. Here $\varphi_{[0,t]}(x) := \{\varphi_t(x) \mid t \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S . Due to the third property we call L the **exit set**.

Definition 6.1.3 (Regular index pairs). We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 6.1.4 (Homotopy equivalence of index pairs). *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S . Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces.*

Remark 6.1.5 (Critical points as isolated compact invariant subsets). Let $f : M \rightarrow \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \rightarrow T_q M$ (after identifying q_i and ∂q_i) where $\varphi(W(q \rightarrow) \cap U) \subseteq T_q^u M$ and $\varphi(W(\rightarrow q) \cap U) \subseteq T_q^s M$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^s M \mid \|v\| \leq \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^u M \mid \|v\| \leq \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \quad (5)$$

Notice that for $k = 0$ we need to consider $\partial D^k = \emptyset$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k :

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{\text{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \quad (6)$$

Remark 6.1.6. Furthermore, this isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{\text{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \quad (7)$$

can be made canonically, using orientations: Let q be a critical point in index λ_q . Let $\mathfrak{b}$ be a ordered basis of $T_q^u M$. Let ψ be a refined Morse chart arround q , such that the induced basis in $T_q M = T_q^u M \oplus T_q^s M$ projects onto a a basis of the first factor $T_q^u M$ that is compatible with $\mathfrak{b}$. Using this Morse chart do define the generator of the regular index pair of q gives a unique generator.

Proof. Let ψ_1 and ψ_2 be two such Morse charts. Notice that they only differ by order of the directions and their signs, meaning that for all i there is a j such that $\psi_1^{-1}(e_i) = \pm \psi_2^{-1}(e_j)$. Hence, viewed as maps into the tangent space we have that $\psi_1(N_p, L_q) = \psi_1(N_p, L_q) \subseteq T_q M$. We define $B_\varepsilon^{\lambda_q}(0) \subseteq \mathbb{R}^{\lambda_q}$ to be the sphere. The generator of $\psi_1(N_p, L_q)$ comes as the pullback of the canonical bott generator of $K^i(B_\varepsilon^{\lambda_q}(0), \partial B_\varepsilon^{\lambda_q}(0))$ via the basis map after the collapse map $c : (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Let α be a generator of $\psi_1(N_p, L_q)$ induced from the basis $\mathfrak{b}_1$ that comes from ψ_1 . I.e.

$$\alpha = (B_{\mathfrak{b}_1} \circ c)^*(\beta)$$

By assumption, the map $B_{\mathfrak{b}_1} \circ B_{\mathfrak{b}}^{-1}$ has determinant one. Hence it is homotopic to the identity, and this homotopy can be reduced to a well defined homotopy on the quotient space $(D_\varepsilon^u / \partial D_\varepsilon^u)$. Hence

$$\alpha = (B_{\mathfrak{b}} \circ c)^*(\beta)$$

And thereby, the generacd tor only depends on $\mathfrak{b}$ concluding the proof. $\square$

Example 6.1.7 (A Map of Homology Groups). Let $f : M \rightarrow \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$ and $p \in \text{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \rightarrow q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0)

is a pair for p . Assuming we have regular index pairs (N_q, L_q) and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \rightarrow p) : K^{k-1}(N_p, L_p) \rightarrow K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \text{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \rightarrow \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q) \quad (8)$$

Definition 6.1.8 (Regular index pairs and a filtration on M). As always let $f : M \rightarrow \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m . Let $\varphi_t : M \rightarrow M$ be the flow given by $-\text{grad} f$. For $0 \leq j \leq k \leq m$ define:

$$W(k, j) = \bigcup_{j \leq \lambda_p \leq \lambda_q \leq k} W(q \rightarrow p).$$

We know that this space is compact. Assume that N is a compact neighbourhood of $W(k, j)$, such that $\text{Crit}(f) \cap N = \text{Crit}(f) \cap W(k, j)$, meaning that N does not contain any new critical points. Then the biggest invariant subspace from definition 6.1.2 is

$$I(N) := \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} = W(j, k).$$

This follows from every gradient flow line starting and ending in a critical point. Therefore, $W(k, j)$ is an isolated compact invariant set. By a corollary of the lambda lemma we know that

$$W_j^s := \bigcup_{j \leq \lambda_p} W(\rightarrow p)$$

$$W_j^u := \bigcup_{\lambda_p \leq j} W(p \rightarrow)$$

for all $j = 0, \dots, m$ are compact as they are finite unions of compact sets. Now we define $N_m := M$ and choose a cofibered compact neighbourhood N_{m-1} of $W_{m-1}^{p \rightarrow}$ that is positively invariant and satisfies $N_{m-1} \cap W_m^{\rightarrow p} = \emptyset$. One could for example define:

$$N_{m-1} := N_m \setminus \left(\bigcup_{q \in \text{Crit}_m(f)} \overset{\circ}{N}_q \right),$$

where N_q is taken from example 6.1.5. Once we check that N_{m-1} is an exit set with respect to N_m we can conclude that (N_m, N_{m-1}) is a regular index pair for $\text{Crit}_m(f)$. The first statement is clear, since every gradient line starting in $N_m \setminus N_{m-1}$ has to pass

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through N_{m-1} , as they go towards other critical points. This tells us that they have to leave $N_m \setminus N_{m-1}$, which is enclosed by N_{m-1} . The second thing to check can be done by showing that there is a neighbourhood $U \subset N_M$ of N_{m-1} that deforms to N_{m-1} . For this notice that each N_q with $\lambda_q = m$ is homeomorphic to D^m . Now choose a open neighbourhood U_q in the disc of its boundary. By definition this retracts to the boundary with a retraction induced by the flow. And finally define $U := N_{m-1} \cap_{q \in \text{Crit}_m(f)} U_q$. This retracts to N_{m-1} telling us that our index pair is indeed regular.

Now we want to inductively define a filtration: For this we choose a compact cofibered neighborhood N_{m-2} of W_{m-2}^u that is positively invariant in N_{m-1} and has an empty intersection with $W_{m-1}^{\rightarrow p}$. This can be done similar to the construction of N_{m-1} but instead of cutting out neighbourhoods if $q \in \text{Crit}_m(f)$, we can cut out tubular neighborhoods of W_{m-1}^s . By iterating this process we get a filtration:

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M, \quad (9)$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k, j)$ for all $0 \leq j \leq k \leq m$.

6.2 The Spectral sequence

Definition 6.2.1. Let M be a smooth manifold with Morse data and

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 6.1.8. Then we define the Groups

$$E_r^{p,q}(M) := \frac{\ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_{p-1}, N_{p-r})]}{\ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_p, N_{p-r})]}.$$

To define a boundary operator, we proceed as follows: The following diagram commutes (by naturality of the boundary in the triple sequence).

$$\begin{array}{ccc} K^{p+q}(N_{p+r-1}, N_{p-r}) & \xrightarrow{\alpha} & K^{p+q}(N_p, N_{p-r}) \\ \downarrow \delta_{triple}^* & & \downarrow \delta_{triple}^* \\ K^{p+q+1}(N_{p+2r-1}, N_{p+r-1}) & \xrightarrow{\beta} & K^{p+q+1}(N_{p+2r-1}, N_p) \longrightarrow K^{p+q+1}(N_{p+r-1}, N_p) \end{array}$$

Hence, $\beta \circ \delta_{triple}^*$ factors over $K^{p+q}(N_{p+r-1}, N_{p-r}) / \ker \alpha$ and because the bottom row is exact (by the triple sequence) we have get a map

$$K^{p+q}(N_{p+r-1}, N_{p-r}) / \ker \alpha \rightarrow \ker [K_{p+q+1}(N_{p+2r-1}, N_p) \rightarrow K^{p+q+1}(N_{p+r-1}, N_p)]$$

Now since $K^{p+q}(N_{p+r-1}, N_{p-r}) \supseteq \ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_{p-1}, N_{p-r})]$ we can restrict the domain and furthermore compose with the quotient map to get a well-defined map

$$\begin{aligned} d_r^{p,q} : E_r^{p,q} &\rightarrow E_r^{p+r, q-r+1} \\ [x] &\mapsto [\beta \circ \delta_{triple}^*(x)] \end{aligned}$$

Referenz
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Salamon

Corollary 6.2.2 (The first page). We start by analysing the first page

$$\begin{aligned} E_1^{p,q}(M) &:= \frac{\ker \left[K^{p+q}(N_p, N_{p-1}) \rightarrow K^{p+q}(N_{p-1}, N_{p-1}) \right]}{\ker \left[\underbrace{K^{p+q}(N_p, N_{p-1}) \rightarrow K^{p+q}(N_p, N_{p-1})}_{\text{id}} \right]} \cong K^{p+q}(N_p, N_{p-1}) \\ &= \tilde{K}^{p+q}(N_p/N_{p-1}) \end{aligned}$$

Now since (N_p, N_{p-1}) is a regular index pair for $\text{Crit}_p(f)$ we have a flow induced homotopy equivalence:

$$N_p/N_{p-1} \simeq \left(\bigcup_{w \in \text{Crit}_p(f)} D_\varepsilon^u(w) \right) / \left(\bigcup_{w \in \text{Crit}_p(f)} \partial D_\varepsilon^u(w) \right) \cong \bigvee_{w \in \text{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D_\varepsilon^u(w) := \varphi_w^{-1}(\{v \in T_w^u M \mid \|v\| \leq \varepsilon\}),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned} E_1^{p,q} &\cong \tilde{K}^{p+q}(N_p/N_{p-1}) \\ &\cong \bigoplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \end{aligned}$$

In sum we have the chain of horizontal isomorphisms:

$$\begin{array}{ccccccc} E_1^{p,q} & \xrightarrow{\alpha} & \tilde{K}^{p+q}(N_p/N_{p-1}) & \xrightarrow{\beta} & \bigoplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) & \longleftarrow & C^p(M, \mathfrak{A}, \tilde{K}^q(pt)) \\ \downarrow d_1^{p,q} & & \downarrow & & \downarrow & & \downarrow \\ E_1^{p+1,q} & \xrightarrow{\alpha'} & \tilde{K}^{p+1+q}(N_{p+1}/N_p) & \xrightarrow{\beta'} & \bigoplus_{w \in \text{Crit}_{p+1}(f)} \tilde{K}^q(pt) & \longleftarrow & C^{p+1}(M, \mathfrak{A}, \tilde{K}^q(pt)) \end{array}$$

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology?

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by the special Filtration in Morse Theory are comparable:

Theorem 6.2.3. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \dots \subseteq N_{m-1} \subseteq N_m = M$ from (9). The following diagram commutes:*

$$\begin{array}{ccc}
C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) & \xrightarrow{\partial^p} & C^{k+1}(M, \mathfrak{A}, \tilde{K}^l(pt)) \\
\downarrow & & \downarrow \\
\bigoplus_{p \in \text{Crit}_k(f)} K^{k+l}(N_p, L_p) & \xrightarrow{\Delta_{k+l}} & \bigoplus_{q \in \text{Crit}_{k+1}(f)} K^{k+l+1}(N_q, L_q) \\
\downarrow & & \downarrow \\
K^{k+l}(N_k, N_{k-1}) & \xrightarrow{\delta_{triple}} & K^{p+l+1}(N_{k+1}, N_k)
\end{array}$$

Proof Strategy:

We will first build up a diagram that encapsulates the connecting homomorphism from K-theory, which was build as follows: Assume that (X, Y, Z) is a topological pair. Then the first map is induced from the kollaps: $Y \rightarrow Y/Z$. Afterwards we have the connecting homomorphism of the pair:

Then the K-groups of the pointed spaces X/X and $X \cup CY$, where the latter denotes a cone above Y agree. From the latter we can quotient out X , and $X \cup CY/X \cong SY$. Hence we have the triple connecting homomorphism as the composition:

$$K^{-1}(Y, Z) \rightarrow K^{-1}(Y) = K(SY) \cong K(X \cup CY/X) \rightarrow K(X \cup CY) \cong K(X/Y)$$

The triple connecting homomorphism for lower K-Groups is defined exactly the same via the identification $K^{-n}(X, Y) = K(S^n \vee (X, Y))$. For higher K-Groups we apply the natural Bott isomorphism and make use of the identity. By doing this we get the tripple homomorphism

$$\begin{array}{ccccc}
K^n(Y, Z) & \xrightarrow{\delta_{triple}} & K^{n+1}(X, Y) \\
\Downarrow & & \Downarrow \\
K^{-n}(Y, Z) & \xrightarrow{i^*} K^{-n}(Y, p) \xrightarrow{\delta_{double}} K^{-n+1}(X, Y) & \xrightarrow{bott} & K^{-n-1}(X, Y)
\end{array}$$

For the proof of the upper square, we actually proof that the following commutes:

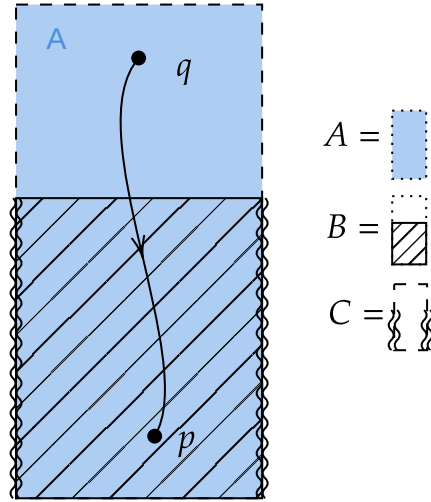
$$\begin{array}{ccc}
C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) & \xrightarrow{\partial^p} & C^{k+1}(M, \mathfrak{A}, \tilde{K}^l(pt)) \\
\downarrow & & \downarrow \\
\bigoplus_{p \in \text{Crit}_k(f)} K^{-k}(N_p, L_p) & \xrightarrow{\Delta_{k+l}} & \bigoplus_{q \in \text{Crit}_{k+1}(f)} K^{-(k+1)}(N_q, L_q)
\end{array}$$

Diagram 10

Furthermore we assumed that $l = 0$. This doesnt change the regularity since if l

was even, we can apply the Bott isomorphism and else the groups are 0. Now we choose a triple (A, B, C) in M such that:

- (a) (A, B) is a regular index pair for a critical point $q \in \text{Crit}_{k+1}$.
- (b) (B, C) is a regular index pair for a critical point $p \in \text{Crit}_k$.



From the perspective of K-theory a regular index pair for a critical point is the same as a sphere with the dimension beeing the index of said critical point. To see this just use a small index pair in a Morse chart (they are all homotopic anyway), collapse the stable part and quotient by the exit set. By doing this we get the commutative diagramm:

$$\begin{array}{ccc}
 K^{k-1}(B, C) & \xrightarrow{\delta_{triple}} & K^k(A, B) \\
 \downarrow & & \downarrow \\
 K^{k-1}(S^{k-1}) & \longrightarrow & K^k(S^{k-1}) \\
 \downarrow & \nearrow & \\
 K^k(S^k) & &
 \end{array}$$

And as a map between K Groups of spheres, we know that this map corresponds to the question of orientation. Now we have the maps

- (a) $S^k \rightarrow S \vee (B/C)$ induced from an orientation in the unstable manifold around p together with $-\text{grad}(f)(x_i)$ where $x_i \in W(q \rightarrow p)$.
- (b) $S^k \rightarrow (A/B)$ induced from an orientation in the unstable manifold around q .

And hence the diagram

$$\begin{array}{ccc} K^k(S \vee (B, C)) & \xrightarrow{\delta_{\text{triple}}} & K^k(A, B) \\ & \nwarrow \quad \nearrow & \\ & K^k(S^k) & \end{array}$$

The map now factors over The K-Group of a bouquet of k -spheres generated over the connecting flow lines. In each such sphere the comparison of orientations happens.

Before we start proving this we need the lemma:

Lemma 6.2.4. *We work in a compact, riemanian, orientable, smooth Manifold (M, g) of dimension m together with a Morse-Smale funktion*

$$f : M \rightarrow \mathbb{R} \quad (11)$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k . we define $f(q) = b$ and $f(p) = a$ and assume that there is no critical point in $f^{-1}(a, b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}, \quad (12)$$

and the constants

$$c \in (a, b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big} . \quad (13)$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}, \quad (14)$$

$$L_q := \{x \in N_q | f(x) = c\}, \quad (15)$$

$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}, \quad (16)$$

$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}, \quad (17)$$

and finally:

$$C := N_p \cup N_q, \quad (18)$$

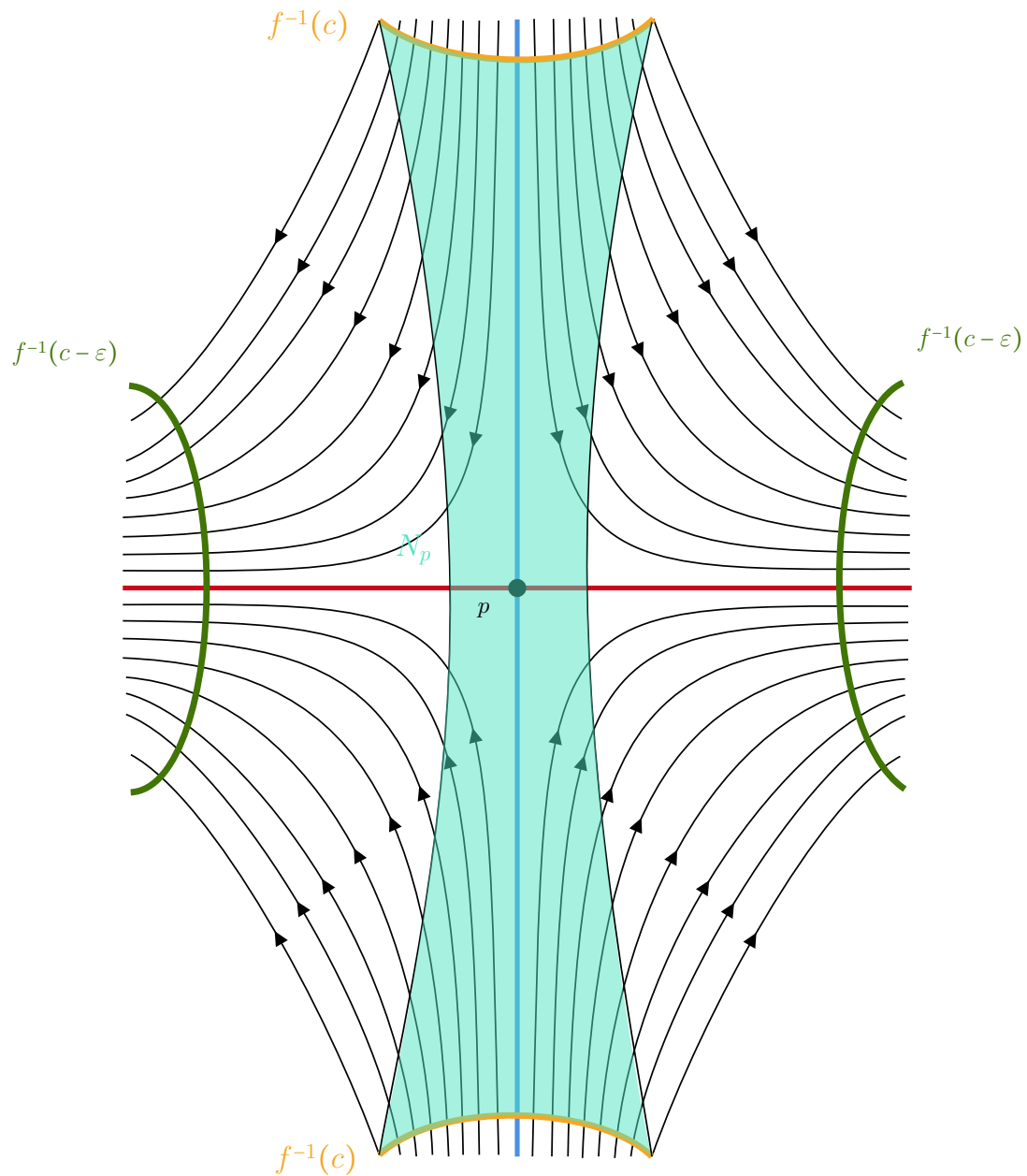
$$B := N_p \cup L_q, \quad (19)$$

$$A := L_p \cup (L_q - \overset{\circ}{N_p}). \quad (20)$$

We claim that

1. (N_q, L_q) is a regular index pair for q .
2. (C, B) is an index pair for q .
3. (N_p, L_p) is a regular index pair for p .
4. (B, A) is an index pair for p .
5. N_p is a tubular neighbourhood of $W(\rightarrow p) \cap M^c$.

The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U . So the only thing left to show is, that N_p is a tubular neighbourhood. in [MorseTheorySalmbo] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [banyaga2004lectures] Banyaga claims this (also without any argument). I find this difficult to proof, since we cannot use the flow for the map from the normal bundle, as points leave N_p along the flow: The Set N_p is sketched in the figure below.



So to prove this, we first formulate two lemmas. First we restate/ refine the λ -lemma:

Lemma 6.2.5 (The Technical λ -Lemma). *Let M be a smooth manifold and let p be a hyperbolic fixpoint of a diffeomorphism φ on M with $\dim W(p \rightarrow) = r$ for $0 < r < m = \dim M$. Let N be an injective immersed submanifold of M that is invariant under φ and has a point of transverse intersection q with $W(\rightarrow p)$. Then for any $\epsilon > 0$, any r -cell neighbourhood D of q in N that also intersects*

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ence
einfügen

$W(\rightarrow p)$ transversal and any r -cell neighbourhood B of p in $W(p \rightarrow)$ there is a smaller cell $\tilde{D} \subset D$, an $n \in \mathbb{N}$ such that the following is true:

We can assume that B and $\varphi^n(\tilde{D})$ lives in a chart centered at p with a product structure $U \cong W(\rightarrow p) \cap B_\mu(0) \times W(p \rightarrow) \cap B_\mu(0)$. Here, the inverse of the projection into $B \subseteq W(\rightarrow p)$:

$$h : B \rightarrow \varphi^n(\tilde{D})$$

satisfies:

1. $\sup_{x \in B} \|x - h(x)\| < \varepsilon$ and
2. For the inclusions $i_1 : B \rightarrow M$ and $i_2 : \varphi^n(\tilde{D}) \rightarrow M$ we have that

$$\sup_{x \in B} \|d(i_1)_x - d(i_2 \circ h)_x\| < \varepsilon.$$

Proof. The proof is essentially the same as given in David Hurtubise - Lectures on Morse Homology. I omitted it here. $\square$

Next we prove that fiberwise the flow is monotonically decreasing in any normal-direction with respect to $W(\rightarrow p)$:

Lemma 6.2.6. *Let U be a tubular neighbourhood of $W(\rightarrow p) \cap M^c$, with $J : NW(\rightarrow p) \cap M^c \rightarrow M$ being the corresponding embedding. For $x \in W(\rightarrow p) \cap M^c$ we denote U_x the embedded fiber over x . Then there exists a $T > 0$ big enough and a global trivialization $\Phi : NW(\rightarrow p) \cap M^c \rightarrow (W(\rightarrow p) \cap M^c) \times \mathbb{R}^{\lambda_p}$ such that the composition $f \circ \varphi_T \circ J \circ \Phi^{-1}$ is fibrewise monotonically decreasing meaning that for any $0 \neq v \in U_x$ and $\nu > 0$ small the function*

$$l_\nu : (1 - \nu, 1 + \nu) \rightarrow \mathbb{R}$$

$$\lambda \mapsto f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda \cdot v)$$

has negative derivative.

Proof. We will apply the technical λ -lemma. For this we reuse the notation as follows. The lemma tells us, that for any $\mu_T > 0$ we can find a T big enough, such that $\varphi_T(J(U_x))$ is close in the following sense: There is a neighbourhood L of p and a Morse chart

$$\Psi : L \rightarrow \mathbb{R}^{\lambda_p} \times \mathbb{R}^{m-\lambda_p}$$

such that $\Psi|_{\mathbb{R}^{\lambda_p}} \supseteq W(p \rightarrow)$. In this chart the set $\varphi_T(J(U_x))$ is the graph of a function $h_x^s : \mathbb{R}^{\lambda_p} \rightarrow \mathbb{R}^{m-\lambda_p}$ that satisfies:

$$\|dh_v^s\| < \mu_T$$

$$\|h_v^s\| < \mu_T$$

We define the map $h_x = h_x^s \oplus \text{id}_{\lambda_p} : \mathbb{R}^{\lambda_p} \rightarrow \mathbb{R}^m$.

Now $\varphi_T(J(U_x)) = \Gamma(h_x^s) = \text{im}(h_x)$. Using this we define our global trivialisation:

$$\begin{aligned} \Phi : NW(\rightarrow p) \cap M^c &\rightarrow (W(\rightarrow p) \cap M^c) \times \mathbb{R}^{\lambda_p} \\ \xi &\mapsto \left(\pi(\xi), h_{\pi(\xi)}^{-1} \circ \Psi \circ \varphi_T \circ J(\xi) \right) \end{aligned}$$

In this trivialization the funktion l_v looks like:

$$\begin{aligned} l_v(\lambda) &= f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda v) \\ &= f \circ \varphi_T \circ J \circ J^{-1} \circ \varphi_{-T} \circ \Psi^{-1} \circ h_x \\ &= f \circ \Psi^{-1} \circ h_x. \end{aligned}$$

In coordinates this reads:

Now we can calculate its derivative to conclude the statement:

$$\begin{aligned} d(l_v)(\lambda) &= d(f \circ \psi^{-1})|_{h_x(\lambda v)} \circ dh_x|_{\lambda v}(v) \\ &= d(f \circ \psi^{-1})|_{h_x(\lambda v)}(v, dh_x^s(v)) \\ &= -2\lambda \sum_{i_1}^{\lambda_p} (v^{i_1})^2 + (h_x^s(\lambda v))^T \cdot (dh_x^s)^i(\lambda v) \\ &\leq -2\lambda \|v\|^2 + \lambda^2 (\mu^T)^2 \|v\|^2 \\ &= (-2 + (\mu^T)^2 \lambda) \lambda \|v\|^2 < 0 \text{ for all } v \neq 0, \lambda \in I \text{ and for all } T \text{ big enough.} \end{aligned}$$

□

Hence, the sets $N_p \cap U_x$ are starshaped with respect to the embedding of the normal space.

Now we want to **smoothly** rescale to the starshaped neighbourhood. This is in general not possible and homeomorphisms from starshaped neighbourhoods to R^n are hard to describe, even if they are not required to be a rescaling. The main problem in the general case is, that the following: for each $v \in \varphi_T(J(U_x))$ by the star shaped property there is a unique λ_v , such that $f \circ (\lambda_v v) = a - \varepsilon$. But the map $v \mapsto \lambda_v$ needs not to be continuous in the general starshaped setting however here it is and it even continuously depends on x :

Definition 6.2.7 (Hyperspherical coordinates). For our next proof we need to intro-

duces spherical coordinates on R^{λ_p} . Define $s_k := \sin(\theta_k)$ and $c_k := \cos(\theta_k)$:

$$\Lambda : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}^{\lambda_p}$$

$$(r, \theta_1, \dots, \theta_{\lambda_p-1}) \mapsto \begin{pmatrix} rc_1 \\ rs_1c_2 \\ \vdots \\ rs_1 \cdots s_{\lambda_p-2}c_{\lambda_p-1} \\ rs_1 \cdots s_{\lambda_p-2}s_{\lambda_p-1} \end{pmatrix}$$

With differential

$$\begin{pmatrix} c_1 & -rs_1 & 0 & 0 & \cdots & 0 \\ s_1c_2 & rc_1c_2 & -rs_1s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ s_1 \cdots s_{n-2}c_{n-1} & \cdots & \cdots & \cdots & -rs_1 \cdots s_{n-2}s_{n-1} \\ s_1 \cdots s_{n-2}s_{n-1} & rc_1 \cdots s_{n-1} & \cdots & \cdots & rs_1 \cdots s_{n-2}c_{n-1} \end{pmatrix}$$

We have shown in lemma 6.2.6 that the maps l_v are monotonically decreasing. Hence the condition $f(x) \geq a - \varepsilon$ gives rise to a starshaped fiber. But we want the fibers to be homeomorphic to R^{λ_p} and the dream would be to just rescale it as follows: Assume that $\rho > 0$ is small enough, such that for all $x \in (W(\rightarrow p) \cap M^c)$ we have that

$\sup_{x \in h_x(B_\rho(0))} (f(x)) \geq a - \varepsilon$. Now we have the maps:

$$r_x : h_x(\partial(B_\rho(0))) \rightarrow \mathbb{R}_{\geq 1}$$

$$v \mapsto \lambda_v \text{ such that } f(\lambda_v v) = a - \varepsilon$$

Notice that for small enough ε such a λ_v exists and the lemma 6.2.6 gives us uniqueness. Now we want to have continouity and furthermore continuous dependence on x for which we will use the parameter-dependent implicit function theorem:

Lemma 6.2.8. *The parameterdependent funktions r_x are continouos and depend continouos on x*

Proof. We redefine the funktion r_x by using spherical coordinates as follows: Implicitly for each x we have the Mapping:

$$F_x : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}$$

$$(r, \theta_1, \dots, \theta_{\lambda_p-1}) \mapsto f \circ h_x \circ \Lambda(r, \theta_1, \dots, \theta_{\lambda_p-1})$$

f mit $f \circ \Psi^{-1}$ ersetzen, da hier eigentlich die Morse karte genutzt wird!!

Here the funktion reads:

$$f \circ h_x \circ \Lambda(\underbrace{r, \theta_1, \dots, \theta_{\lambda_p-1}}_{:=v}) = f\left(\begin{pmatrix} h_x^s(\Lambda(v)) \\ \Lambda(v) \end{pmatrix}\right) = -2r^2 + 2 \sum_{i=\lambda_p+1}^m (h_x(\Lambda(v)))_i^2$$

To make use of the implizit funktion theorem, we inspect the differential in r -direktion:

$$\frac{\partial}{\partial r}(-2r^2 + 2 \sum_{i=\lambda_p+1}^m (h_x(\Lambda(v)))_i^2) = -4r + 2 \sum_{i=\lambda_p+1}^m \left(\sum_{j=1}^{\lambda_p} \left(\frac{\partial(h_x)_i}{\partial x_j} \frac{\partial \Lambda_j}{\partial r} \right) \right)$$

This does not vanish for big enough r , since the partial differentials of λ are of norm one, and those of h_x are small. Hence for big enough r , lets say $r \geq \mu > 0$ this is negative and thereby we get a funktion

$$\tilde{r}_x : [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}_{\geq \mu}$$

such that $(f \circ h_x \circ \Lambda)^{-1}(a - \varepsilon)$ is the graph of $\tilde{r}_x$. Hence, if we define $\pi_r : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow [0, \pi]^{\lambda_p-2} \times [0, 2\pi)$ to be the projection we have that

$$\begin{aligned} r_x : h_x(\partial B_\rho(0)) &\rightarrow \mathbb{R} \\ v &\mapsto \tilde{r}_x \circ \pi_r(\Lambda^{-1}(h_x^{-1}v)) \end{aligned}$$

is continouos and depends continouosly on x . (In reality we still need to check for smoothness in $\Lambda(\{\theta_{\lambda_p-1} = 0\})$. This can be done by a different parametrisation of the sphere meaning by different spherical coordinates.) The resulting funktion agrees with the earlier definition of r_x concluding the proof. $\square$

Proof That N_p is Tubular. To finish our proof we rescale the Fibers smoothly. For this we first restrict the domain of J : Let $\nu > 0$ be small enough, such that for

$$N_\nu W(\rightarrow p) \cap M^c := \{\xi \mid \text{if } \Phi(\xi) = (x, v), \|v\| < \nu\}$$

the supremum

$$\sup_{x \in \varphi_T \circ J(N_\nu W(\rightarrow p) \cap M^c)} \geq a - \varepsilon$$

Hence, $J(N_\nu W(\rightarrow p) \cap M^c)$ is included in N_p and now we resscale the fibres $U_x^\nu := U_x \cap \text{im} J(N_\nu W(\rightarrow p) \cap M^c) \supseteq U_x$ to get a surjective inclusion into N_p . For all $x \in W(\rightarrow p) \cap M^c$ we define the smooth rescaling as follows:

$$\begin{aligned} RS : U_x^\nu &\rightarrow U_x \\ v &\mapsto \varphi_{-T} \circ \Psi^{-1} \circ r_x \left(\rho \cdot \frac{h_x^{-1} \circ \Psi \circ \varphi_T(v)}{\|h_x^{-1} \circ \Psi \circ \varphi_T(v)\|} \right) \end{aligned}$$

This rescaling is smooth, smoothly depends on x , and it maps surjectifly onto $N_p \cap U_x$. $\square$

The same arguments work to show that N_q is a tubular neighbourhood of $W(q \rightarrow) \cap M_c$, by replacing f with $-f$. Furthermore, we can assume, that the V_j from below are fibers in the normal neighbourhood. This can be realized, since V_j intersect transversal and by trivialization we have a global fram of the bundel and hence can smoothly rescale the embedding along the frame to get that V_j is the image of the fibre above x_j .

Commuting triple connecting homomorphisms. We first notice, that we have the map of triples given by the inclusion:

$$\left(W(q \rightarrow) \cap N_q, S(q \rightarrow), S(q \rightarrow) \setminus \bigcup_j V_j \right) \rightarrow (C, B, A)$$

and hence by naturality we have the diagram, where the top square commutes:

$$\begin{array}{ccc} K^{-\lambda_p} [B, A] & \xrightarrow{\delta_{triple}^1} & K^{-\lambda_q} [C, B] \\ \downarrow i_{B,A}^* & & \downarrow i_{C,B}^* \\ K^{-\lambda_p} \left[S(q \rightarrow), S(q \rightarrow) \setminus \bigcup_j \text{int}(V_j) \right] & \xrightarrow{\delta_{triple}^2} & K^{-\lambda_q} [W(q \rightarrow) \cap N_p, S(q \rightarrow)] \\ \downarrow i_j^* \uparrow c_j^* & \nearrow \delta_{triple}^j & \\ K^{-\lambda_p} [V_j, \partial V_j] & & \end{array}$$

Diagram 21

Furthermore, in this diagram the tuple $\left(K^{-\lambda_p} \left[S(q \rightarrow), S(q \rightarrow) \setminus \bigcup_j \text{int}(V_j) \right], (c_j^*)_j \right)$ is a coproduct, and we define $\delta_{triple}^j := \delta_{triple}^2 \circ c_j^*$. Then we have that

$$\delta_{triple}^2 = \sum_j \delta_{triple}^j \circ i_j^*$$

The map δ_{triple}^2 is defined as the composition:

$$\begin{array}{ccccc} K^{-\lambda_p} [V_j, \partial V_j] & & & & \\ \downarrow c_j^* & & & & \\ K^{-\lambda_p} \left[S(q \rightarrow), S(q \rightarrow) \setminus \bigcup_j \text{int}(V_j) \right] & \xrightarrow{\delta_{triple}^2} & K^{-\lambda_q} [W(q \rightarrow) \cap N_p, S(q \rightarrow)] & & \\ \downarrow i_j^* & & \uparrow \beta \circ \text{triv}_{C_1 S(q \rightarrow)} & & \\ K^{-\lambda_p} [S(q \rightarrow)] & \xrightarrow{h_1^1} K^{-\lambda_p+1} [S_1 \vee S(q \rightarrow)] \xrightarrow{h_2^2} K^{-\lambda_p+1} [W(q \rightarrow) \cap N_p \cup C_1(S(q \rightarrow)), W(q \rightarrow) \cap N_p] \xrightarrow{i_2^2} K^{-\lambda_p+1} [W(q \rightarrow) \cap N_p \cup C_1(S(q \rightarrow))] \end{array}$$

Diagram 22

The downwards maps in the Theorem are the induction of the canonical generators. This is done by the following assignement

Now the composition

$$h_1 * i_1 * \circ c_j^* : K^{-\lambda_p} [V_j, \partial V_j] \rightarrow K^{-\lambda_p+1} [S_1 \vee S(q \rightarrow)]$$

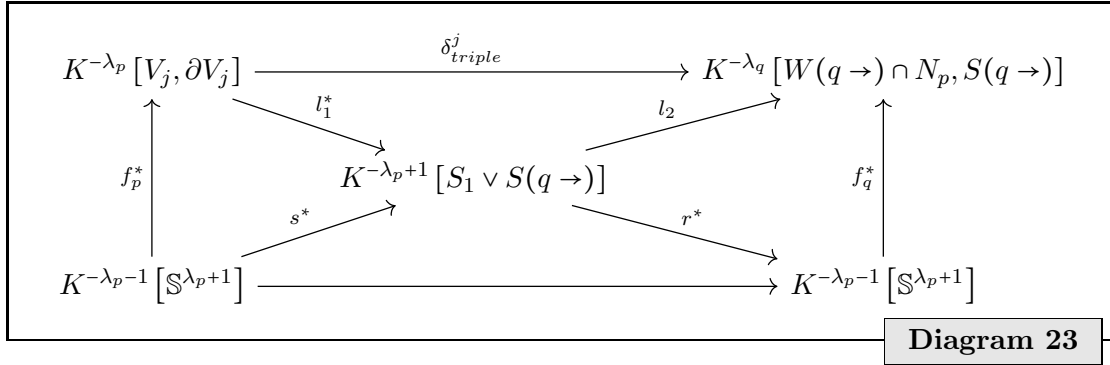
can be simplified by the map

$$l^* : K^{-\lambda_p} [V_j, \partial V_j] = K^{-\lambda_p+1} [S_1 \vee (V_j, \partial V_j)] \rightarrow K^{-\lambda_p+1} [S_1 \vee S(q \rightarrow)]$$

where l is the collapsing map.

$$\begin{aligned} l : S_1 \vee S(q \rightarrow) &\rightarrow S_1 \vee (V_j, \partial V_j) \\ (t, x) &\mapsto (t, \bar{x}) \end{aligned}$$

So now we are in the following situation:



Now we have a list of the Functions from the diagram above:

- The map f_p induces α_p . Remember, that α_p comes from the canonical generator α'_p of the index pair $[B, A]$ and α'_p is defined as

$(\varphi \circ \tilde{\psi}^{-1})^*(\beta)$ where φ is the homotopie equivalence of index pairs

$\tilde{\psi}$ is a Morse chart adapted to the basis of $T_p^u M$,

and β is the bott generator of the sphere $[D_\varepsilon^u \times D_\varepsilon^s / \partial D_\varepsilon^u \times D_\varepsilon^s]$.

Instead of using φ we can rescale $\tilde{\psi}$ to be a mapping to $[B, A]$ instead of first mapping to the prototype index pair $[D_\varepsilon^u \times D_\varepsilon^s, \partial D_\varepsilon^u \times D_\varepsilon^s]$. We call this rescaled Morse chart ψ . Then $\psi^{-1}(\beta) = (\varphi \circ \tilde{\psi}^{-1})^*(\beta)$ since the map $\tilde{\psi} \circ \varphi^{-1} \circ \psi^{-1}$ has the local and hence global degree one. (this can be checkt by differentiatiaion at the origin). Hence $\alpha'_p = (\psi^{-1})^*(\beta)$. The inclusion

$$i_p : (V_j, \partial V_j) \rightarrow (B, A)$$

induces an isomorphism in the K-groups since N_p is a tubular neighbourhood of the contractible $W(\rightarrow p) \cap M^c$ and the contraction to the fiber V_j contracts L_p to ∂V_j (the contraction is well defined on the quotients, and for K-Theory, we only care for the qutoient spaces). Now $\alpha''_p = i_p^*(\alpha'_p)$. Define the two funktions

$$J : \mathcal{N}^{W(q \rightarrow)} W(\rightarrow p) \cap M^c \rightarrow N_p \text{ the normal neighbourhood embedding,}$$

$$\nu : \mathcal{N}^{W(q \rightarrow)} W(\rightarrow p) \cap M^c \rightarrow W(\rightarrow p) \times T_p^u M \text{ The trivialization of the normal bundle.}$$

Achtung
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 f_p noch
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oder im
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ändern.

Define $\mathfrak{b}^p$ to be an ordered basis of $T_p^u M$ that is compatible to the orientation of said tanged space. Then assume that J is defined, such that

$$(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1})^* (\beta) = \alpha_p''.$$

If this is not the case just compose J with a basechange of degree one. Now α_p'' generates $K^{-\lambda_p}(V_j, \partial V_j)$. The latter is isomoprhic to $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$ by definition. Now we denote the suspension is the wedge produkt with a sphere. Hence for $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$ we can induce the generator by the function

$$\begin{aligned} f_p : I \times V_j / \partial(I \times V_j) &\rightarrow S \wedge S^{\lambda_p} \simeq S^{\lambda_p+1} \\ (t, x) &\mapsto \Omega(t, \Gamma \circ (B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1})(x)). \end{aligned}$$

Γ maps the disk modulo its boundary to a sphere and Ω maps the suspension of the sphere to the sphere one dimension higher. We abuse the notation a bit and identify the map $(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1})$ with the one defined by quotienting out the boundary. We will leave out the quotienting in the notatino from now on to make the prove easier to read. The invested reader should check, if the quotient maps are always well defined! Now finally we have a little definition issue: In the diagramm we have that

specify
the iso-
morphism
!

$$f_p^* : K^{\lambda_p-1}[\mathbb{S}^{\lambda_p+1}] \rightarrow K^{-\lambda_p}[V_j, \partial V_j],$$

whilst at the moment, f_p maps to $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$. Hence we need to compose this f_p^* with the isomorphism that comes from the suspension axiom:

$$K^{-\lambda_p}(X) \cong K^{-\lambda_p+1}(SX).$$

By abuse of notation we dont write that idetification and rather think of the top left corner as $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$.

- Again, we want f_q to be the map that induces the generator of $K^{-\lambda_q}[W(q \rightarrow) \cap N_q, S(q \rightarrow)]$ which comes from the canonical generator of $[C, B]$ together with the inclusion. Again instead of first introducing the generator on the prototype index pair, we can use E_q^{u-1} instead of the Morse chart. Again we can argue here, that the local degree of changing between the two is one, since E_q^{u-1} is a refined morse chart which can be choosen to be adapted to the ordered basis $\mathfrak{b}^q$ of $T_q^u M$. Hence we can define f_q to be the quotient map:

$$\begin{aligned} W(q \rightarrow) \cap N_q / S(q \rightarrow) &\rightarrow S^{\lambda_q+1} \\ x &\mapsto \Gamma \circ B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1}(x). \end{aligned}$$

Here Γ maps the disk modulo its boundary to the sphere. Hence $f_q = (B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1})^* : K$

- The map l_1 is the collapsing map

$$S_1 \wedge S(q \rightarrow) \rightarrow S_1 \wedge (V_j / \partial V_j)$$

- The map l_2 is not topologically induced.
- The map s is given by $f_p \circ l_1$. Again we abuse the notation and leave out the suspension isomorphism when

$$s^* : K^{-\lambda_p+1}[S_1 \wedge S(q \rightarrow)] \rightarrow K^{-\lambda_p-1}[\mathbb{S}^{\lambda_p+1}].$$

- To find the map r that fits the diagram (to make it commute) is the next goal. First we need to dig into l_2 : From diagram 22 we know that $l_2 = \Sigma \circ \text{triv}_{C_1 S(q \rightarrow)} \circ i_2^* \circ h_2^*$, Where Σ denotes the suspension isomorphism. Hence, if the function r^* makes the diagram commutative we need to ensure, that $f_q^* \circ r^* = l_2$. If we define the function

$$\text{col} : W(q \rightarrow) \cap N_p \cup C_1 S(q \rightarrow) \rightarrow S_1 \wedge S(q \rightarrow)$$

to be the collision, of $W(q \rightarrow) \cap N_p$ to a point, then $i_2^* h_2^* = \text{col}^*$. Now while the function $\text{triv}_{C_1 S(q \rightarrow)}$ is not topologically induced, its inverse is. Lets call the collapse of $C_1 S(q \rightarrow)$ c_1 . Then since Σ is a natural transformation we have

$$f_q^* \circ r^* = l_2 \Leftrightarrow c_1^* \circ f_q^* \circ r^* = \Sigma \circ \text{col}^*$$

Now all maps are topologically induced except Σ and the latter is just used to get the index in the K group right. Hence, if r would be such that

$$r \circ f_q \circ c_1 = \text{col},$$

then the requirement on the induced maps would be satisfied. We first however need to formulate a little lemma:

Lemma 6.2.9. *Let $S^k := \{x \in \mathbb{R}^{k+1} \mid \|x\| = 1\}$, $H_+ := \{x \in S^k \mid x_0 \geq 0\}$ denote the upper hemisphere, H_- the lower hemisphere, $\pi : S^k \rightarrow D^k$ with $\pi(x_0, \dots, x_k) = (x_1, \dots, x_k)$ be the projection and Λ be the homeomorphism from $D^k / \partial D^k \rightarrow S^k$. Let furthermore $c_{Eq} : S^k \rightarrow \overline{H_+} \vee \overline{H_-}$ be the collapse of the equator, where $\overline{H_+}$ denotes the upper hemisphere modulo the equator and $\overline{H_-}$ is defined analog. Let finally $c_2 : \overline{H_+} \vee \overline{H_-} \rightarrow \overline{H_+}$ be the collapse of the second summand and analogously $c_1 : \overline{H_+} \vee \overline{H_-} \rightarrow \overline{H_-}$. Then we have that $\Lambda \circ \pi \circ c_2 \circ c_{Eq} \simeq -\Lambda \circ \pi \circ c_1 \circ c_{Eq}$, where the minus denotes any map homeomorphism of negative degree.*

Proof. This lemma is the concrete version of a standard lemma that is often formulated along the lines: Let $q : S^k \rightarrow S^k \vee S^k$ be the collapse of the equator and p_i be the projection to the i -th summand, then $p_1 \circ q = -p_2 \circ q$. $\square$

Now we need to transport our situation to fit the lemma. For this we will use the maps defined in the lemma above, and need a bunch of maps again:

- First we define the map $\rho' : H_- \rightarrow H_+ \cap H_-$ to be $B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q} \circ (\cdot 3) \circ \pi$ for all x such that $x_0 \neq 1$. Here, the map ρ is given by

$$\begin{aligned} \rho : W(q \rightarrow) \setminus \{q\} &\rightarrow W(q \rightarrow) \cap f^{-1}(c) \setminus W(q \rightarrow) \setminus \{q\} \\ x &\mapsto \varphi_{\tau(x)}(x), \end{aligned}$$

and

$$\begin{aligned} \tau : W(q \rightarrow) \setminus \{q\} &\rightarrow \mathbb{R} \\ x &\mapsto \tau(x) \text{ such that } f \circ \varphi_{\tau(x)}(x) = c. \end{aligned}$$

Furthermore $(\cdot 3)$ denotes the stretching of D^k (i.e. the image of π) to be the ball of diameter three. The north pole gets mapped to any element of the equator. (This choice will not matter later)

- $\Phi : S^{\lambda_q} \rightarrow W(q \rightarrow) \cap N_p \cup C_1 S(q \rightarrow)$ is defined with a case distinction. We define

$$\Phi|_{H_+}(x) = E_q^u \circ B_{\mathfrak{b}^q}^{-1} \circ \pi(x) \quad \text{and} \quad \Phi|_{H_-}(-x_0, \dots, x_{\lambda_q}) = (x_0, f_q^{-1} \circ \rho').$$

Notice, how this function is continuous and does not depend on the choice taken in the definition of ρ' .

- $\Psi : S^{\lambda_q} \vee S^{\lambda_q} \rightarrow (W(q \rightarrow) \cap N_p / S(q \rightarrow)) \vee \mathbb{S} \wedge S(q \rightarrow)$ is given by mapping $x \mapsto \overline{\Phi(x)}$. The well definition is easy to see.
- p_i for $i = 1, 2$ denotes the projection onto the first or second summand in a pointed sum.
- $C_{S(q \rightarrow)} : W(q \rightarrow) \cap N_q \cup C_1 S(q \rightarrow) \rightarrow (W(q \rightarrow) \cap N_p / S(q \rightarrow)) \vee \mathbb{S} \wedge S(q \rightarrow)$ is the collaps of $S(q \rightarrow)$.
- $R : \mathbb{S}^{\lambda_q} \rightarrow \mathbb{S} \wedge S(q \rightarrow)$ is defined to be $col \circ \Phi$

This gives the commutative diagram,

Diagram 24

Now define $r := -R$. For example compose R with the reflection $(x_0, \dots, x_{\lambda_q}) \mapsto (-x_0, \dots, x_{\lambda_q})$. Then r collapses the lower hemissphere. Then by the lemma above and commutativity we have up to homotopy:

$$\begin{aligned} r \circ f_q \circ c_1 &= -R \circ f_q \circ c_1 \circ \Phi \circ \varphi - 1 \\ &= -R \circ \Lambda \circ c_2 \circ C_{Eq} \circ \varphi - 1 \\ &= R \circ \Lambda \circ c_1 \circ C_{Eq} \circ \varphi - 1 \\ &= col. \end{aligned}$$

With r we finally have all the maps. Now we want to find out what degree the composition $s \circ r$ has. For this we calculate the local degree in some point by checking the sign of the determinant of the Jacobean.

Now let $\tilde{y} = \frac{1}{3}(B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1}(x_j))$, and $y := \Gamma(y)$. The multiplication by three in the definition made sure that y is in the interior (and in fact in the upper hemisphere). Now $\pi_+ : H_+ \rightarrow D^{\lambda_q}$ is a chart for the upper hemisphere that is the same orientation as $-\pi_- : H_- \rightarrow D^{\lambda_q}$. Around y the function is a diffeomorphism, hence we can check for local degree by checking the sign of the determinant of the Jacobean of $s \circ r$ in y with respect to charts around y and around its image that are induced from the same orientation in S^k . Since y gets mapped to the lower hemisphere we check the sign of the jacobian of

$$-\pi_- \circ s \circ r \circ \pi_+^{-1} \quad \text{in } x, \text{ where } x := \pi_+(y)$$

So lets expand this definition:

$$\begin{aligned}
-\pi_- \circ s \circ r \circ \pi_+^{-1}(x) &= -\pi_- \circ s \circ col \circ \Phi \left(\underbrace{-\left(\pi_+^{-1}(x)\right)_0}_{:=t}, \left(\pi_+^{-1}(x)\right)_{i=1,\dots,\lambda_q} \right) \\
&= -\pi_- \circ s \circ col \left(-t, f_q^{-1} \circ \rho' \left(\left(\pi_+^{-1}(x)\right)_{i=1,\dots,\lambda_q} \right) \right) \\
&= -\pi_- \circ f_p \circ \underbrace{\circ l_1 \circ col}_{=id \text{ around } y} \left(-t, f_q \circ \rho' \left(\left(\pi_+^{-1}(x)\right)_{i=1,\dots,\lambda_q} \right) \right) \\
&= -\pi_- \circ f_p \left(-t, f_q^{-1} \circ \rho' \left(\left(\pi_+^{-1}(x)\right)_{i=1,\dots,\lambda_q} \right) \right) \\
&= -\pi_- \circ \Omega \circ \left(-t, \Gamma \circ B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ f_q^{-1} \circ \underbrace{B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q}(3 \cdot x)}_{=\rho \circ \pi_+^{-1}(x)} \right) \\
&= -\pi_- \circ \Omega \left(-t, \Gamma \circ \underbrace{B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q}(3 \cdot x)}_{=:z} \right) \\
&= -\pi_- \circ \Omega_{\lambda_q} \circ \left(-t, \Omega_{\lambda_p} \left(\|z\|, \frac{z}{\|z\|} \right) \right) \\
&= -\pi_- \circ \Omega_{\lambda_q} \circ (-t, (-\cos(\pi \|z\|), \sin(\pi \|z\|) z)) \\
&= \sin(-\pi t) (-\cos(\pi \|z\|), \sin(\pi \|z\|) z)
\end{aligned}$$

We now want to look at the differential of this in x . To be specific we only care for the sign of the determinant of said differential. Since $t \in (-\frac{1}{2}, 0)$ is negative, the scalar multiplication with $\sin(-\pi t)$ has a jacobian with a positive determinant. In Fact, its jacobian is Hence it does not interfere with the sign and we can leave it out. Hence we care for the sign of the jacobian of the map

should be
 $-\frac{1}{3}$

$$x \mapsto (-\cos(\pi \|z\|), \sin(\pi \|z\|) z) .$$

Now the jacobian of the map splitts into two block matrixies: Tut sie nicht!! hier sauber reingehen! vielleicht auch mal $\|(\|z\|)$ ableiten...

$$\begin{pmatrix} \sin(\pi \|z\|) & 0 \\ 0 & B \end{pmatrix}$$

For this to be true, we need $\mathfrak{b}^q = (\mathfrak{b}_i^q)_{i=1,\dots,\lambda_q}$ to be a basis such that the following is true: Let Q be the map from $T_q^u M \rightarrow T_{E_q^{u-1}(x_j)} T_q^u M$. Then we want that $Q(\partial(\mathfrak{b}_1^q))$ maps to $-\text{grad}_{x_j}(f)$ via $dE_q^u - 1$. Furthermore, all other $\mathfrak{b}_i^q$ get mapped to something orthogonal. Then we have the following two lemmas to help us compute:

Theorem 6.2.10. *The Map*

$$\tau : W(q \rightarrow) \setminus \{q\} \rightarrow \mathbb{R} \quad (25)$$

$$x \mapsto \tau(x) \text{ such that } f \circ \varphi_{\tau(x)}(x) = c \quad (26)$$

is smooth.

Proof. We know that such a map exists and is well defined since every flow line passes through $f^{-1}(c)$. Now let $x \in W(q \rightarrow) \setminus \{q\}$, V be a neighbourhood of x and U be a neighbourhood of $\varphi_{\tau(x)}(x)$. Define the map

$$\begin{aligned} G : V \times I &\rightarrow \mathbb{R} \\ (x, t) &\mapsto f \circ (\varphi_t(x)) . \end{aligned}$$

Now the partial derivative reads

$$\frac{\partial}{\partial t} G|_{x, t} = df|_{\varphi_t(x)} \circ \frac{\partial}{\partial t} \varphi_t(x)|_t = df|_{\varphi_t(x)} \circ (-\text{grad}_{\varphi_t(x)}(f)) .$$

For $\varphi_{\tau(x)}$ we have that $T_{\varphi_{\tau(x)}(x)} f^{-1}(c) = \ker df|_{\varphi_{\tau(x)}(x)}$. Since the flow line is transversal to the level set or said differently, the gradient is orthogonal to level sets, we have that the latter does not live in the kernel and hence

$$\frac{\partial}{\partial t} G|_{x, \tau(x)} = df|_{\varphi_{\tau(x)}(x)} \left(-\text{grad}_{\varphi_{\tau(x)}(x)}(f) \right) \neq 0 .$$

This lets us use the implicit function theorem to deduce that the level set $G = c$ is locally the graph of a function $\tau : U_x \rightarrow \mathbb{R}$ where U_x is a neighbourhood of x . Hence the function τ is globally smooth since this argument can be done everywhere. $\square$

Remark 6.2.11. Now we care for the differential of τ in a point $x_j \in W(q \rightarrow) \cap f^{-1}(c)$. For this we choose a chart of $W(q \rightarrow)$ around x_j such that the induced bases of $T_{x_j} W(q \rightarrow) = \langle -\text{grad}_{x_j}(f) \rangle \oplus T_{x_j} V_j$ that respects the splitting. for example $\frac{\partial}{\partial x^1}|_{x_j} = -\text{grad}_{x_j}(f)$ and $\frac{\partial}{\partial x^i}|_{x_j} \in T_{x_j} V_j$ for i bigger than one. With this we can start our calculations: Notice how $x \mapsto f \circ \varphi_{\tau(x)}(x)$ is a constant function and hence its differential is 0.

$$\begin{aligned} 0 = d_x (f \circ \varphi_{\tau}) &= \overbrace{df}^{\text{1-form}} (d\varphi_{\tau}) = df \left(d_t \varphi_t \circ d_x \tau(x) + d_x \varphi_{\tilde{t}=\tau(x)} \right) \\ &= df (d_t \varphi_t \circ d_x \tau(x)) + df \left(d_x \varphi_{\tilde{t}=\tau(x)} \right) \\ &= d_t (f \circ \varphi_t) \circ d_x \tau(x) + df \left(d_x \varphi_{\tilde{t}=\tau(x)} \right) \end{aligned}$$

Hence we know that

$$d_x \tau = - \frac{df \left(d_x \varphi_{\tilde{t}=\tau(x)} \right)}{d_t (f \circ \varphi_t)} = - \frac{df \left(d_x \varphi_{\tilde{t}=\tau(x)} \right)}{-\|\text{grad}(f)^2\|}$$

Now we can calculate in x_j where $\tau(x_j) = 0$

$$d_x \tau|_{x_j} \frac{\partial}{\partial x^i}|_{x_j} = \frac{df|_{x_j} \frac{\partial}{\partial x^i}|_{x_j}}{\|\text{grad}(f)^2\|} = \frac{g(\text{grad}_{x_i}(f), \frac{\partial}{\partial x^i}|_{x_j})}{\|\text{grad}(f)^2\|} = \begin{cases} -1 & \text{if } i = 1 \\ 0 & \text{else.} \end{cases}$$

Theorem 6.2.12. *The map*

$$\begin{aligned} \rho : W(q \rightarrow) \setminus \{q\} &\rightarrow W(q \rightarrow) \cap f^{-1}(c) \setminus W(q \rightarrow) \setminus \{q\} \\ x &\mapsto \varphi_{\tau(x)}(x). \end{aligned}$$

is a submersion in $y \in S(q \rightarrow)$

find a chart around x_j such that the induced chart respects the splitting!

Proof. To prove this we calculate its differential $d\rho|_y$ as follows:

$$d\rho|_y = d_x(\varphi_\tau)|_y = d_t\varphi_t|_{t=\tau(x)} \circ d_x\tau|_y + d_x\varphi_{\tau(y)}|_y$$

We can rewrite this by viewing $d_x\tau|_y$ as a one form and using the definition of the flow:

$$d_t\varphi_t|_{t=\tau(x)} = -\text{grad}_{\rho(x)}(f):$$

$$d\rho|_y = -\text{grad}_{\rho(x)}(f) \cdot d_x\tau|_y + d_x\varphi_{\tau(y)}|_y$$

Now if $y \in f^{-1}(c)$ we have that $\tau(y) = 0$ and hence $d_x\varphi_{\tau(y)} = \text{id}$. Hence the differential reads:

$$d\rho|_y = -\text{grad}_y(f) \cdot d_x\tau|_y + \text{id}.$$

Now we can choose a local chart around y such that $-\text{grad}_y(f)$ becomes the first basis vector of the tangend space, and the rest are orthonormal. Then the differential acts on $-\text{grad}_y(f)$:

$$d\rho|_y(-\text{grad}_y(f)) = -\text{grad}_y(f) \cdot \underbrace{d_x\tau|_y(-\text{grad}_y(f))}_{=-1} - \text{grad}_y(f) = 0$$

and on all the other tangend vectors as the identity, due to the first summand vanishing. Therefore the jacobian in this basis is of the form

$$d\rho|_y = (0, 1_{\lambda_q-1}).$$

□

□

Back to our differential: We now have that

$$d(\rho \circ E_q^u \circ B_{\mathfrak{b}^q}(3 \cdot x))|_y(e_i) = \begin{cases} 0 & \text{if } i=1 \\ 3\mathfrak{b}_i^q & \text{else.} \end{cases}$$

For the further calculation we claim that $\|(\|z) = 1 - \tau(3x)$.